



FINAL MARK

**GIRRAWEEEN HIGH SCHOOL
Mathematics Extension 1
HSC ASSESSMENT Task 2, 2016
ANSWERS COVER SHEET**

Name:

QUESTION	MARK	HE2	HE3	HE4	HE5	HE6	HE7
Q1 - Q5	/5						✓
Q6	/19						✓
Q7	/23						✓
Q8a	/4	✓					✓
Q8bc	/11		✓				✓
Q8 Total	/15						✓
Q9	/19						✓
Q10	/12						✓
TOTAL							
	/93	/4	/11				/93

HSC Outcomes

Mathematics Extension 1

- HE2 uses inductive reasoning in the construction of proofs.
- HE3 uses a variety of strategies to investigate mathematical models of situations involving binomial probability, projectiles, simple harmonic motion and exponential growth and decay.
- HE4 uses the relationship between functions, inverse functions and their derivatives
- HE5 applies the chain rule to problems including those involving velocity and acceleration as functions of displacement.
- HE6 determines integrals by reduction to a standard form through a given substitution.
- HE7 evaluates mathematical solutions to problems and communicates them in an appropriate form.



GIRRAWEEEN HIGH SCHOOL

YEAR 12 HALF YEARLY EXAMINATION

2016

MATHEMATICS EXTENSION 1

Time Allowed: Two hours

(Plus 5 minutes reading time)

Instructions To Students

- **Attempt all questions.**
- **All necessary working must be shown for questions 6-10.**
- **Marks may be deducted for careless or badly arranged work.**
- **Board approved calculators may be used.**
- **For Questions 1-5, write the letter corresponding to the correct answer on your answer sheet.**
- **For Questions 6-10, start each question on a new sheet of paper. Each question should be clearly labelled.**
- **Write 'End of Solutions' on your answer paper when you finish answering all questions.**

For Questions 1-5, write the letter corresponding to the correct answer on your answer sheet (5 marks)

1. What is the solution to the inequality $3 - x \geq \frac{2}{x}$.
(A) $x < 0$ or $1 \leq x \leq 2$
(B) $x \geq 2$ or $0 < x \leq 1$
(C) $x > 0$ or $-2 \leq x \leq -1$
(D) $x \leq -2$ or $-1 \leq x < 0$
2. What is the acute angle between the lines $x - y + 2 = 0$ and $2x - y - 1 = 0$.
(A) $18^\circ 26'$ (B) $19^\circ 28'$ (C) $70^\circ 32'$ (D) $71^\circ 32'$
3. Twelve people sit at a round table. The number of arrangements possible if two particular persons are seated together is
(A) 24 (B) 3 628 800 (C) 725 760 (D) 7 257 600
4. The coefficient of x^6 in the expansion of $\left(\frac{1}{x^2} - x\right)^{18}$.
(A) 18 564 (B) 3060 (C) 43 758 (D) -3060
5. A curve has parametric equations $x = \frac{2}{t}$ and $y = 2t^2$. What is the Cartesian equation of this curve.
(A) $y = \frac{4}{x}$ (B) $y = \frac{8}{x}$ (C) $y = \frac{4}{x^2}$ (D) $y = \frac{8}{x^2}$

Question 6 (19 marks)

- (a) Solve $\frac{2x}{x+1} \leq 1$ 3
- (b) Find the coordinates of the point P which divides the interval joining $A(-4, -6)$ and $B(6, -1)$ externally in the ratio 3:2. 3
- (c) Solve: $\log_e(10x + 24) = 2\log_e x$ 4
- (d) (i) Express $\sin x + \sqrt{3} \cos x$ as $A \sin(x + \alpha)$. 3
- (ii) Hence solve $\sin x + \sqrt{3} \cos x = \sqrt{2}$, $0 < \alpha < \frac{\pi}{2}$, $0 \leq x \leq 2\pi$. 2
- (e) The gradient of the tangent at any point on a curve is given by $\frac{dy}{dx} = e^{-2x}$. If the curve passes through the point $(0, 1)$
- (i) Find the equation of the curve. 3
- (ii) The point on the curve with x -coordinate $-\frac{1}{2} \log_e 3$. 1

Question 7 (23 marks)

(a) Differentiate:

(i) $y = x^2 \log_e(x^2)$ (ii) $y = \frac{e^x}{x^3}$ (iii) $y = \log_e \left(\frac{x}{\sqrt{x^2 + 1}} \right)$ 10

(b) Find:

(i) $\int x^2 e^{x^3+1} dx$ (ii) $\int \frac{3x}{x^2+11} dx$ (iii) $\int_1^{\log_e 2} \frac{e^{3x} + 1}{e^x} dx$ 9

(c)(i) Find the derivative of $\log_e(x^2 - 5x + 7)$. 2

(ii) Hence evaluate $\int_3^4 \frac{(2x-5)}{x^2-5x+7} dx$ correct to 4 significant figures. 2

Question 8 (15 marks)

(a) Use Mathematical Induction to prove that $\sum_{r=1}^n r \times r! = (n+1)! - 1$ for $n \geq 1$. 4

(b) Expand $\left(2 - \frac{1}{x}\right)^3$ and hence determine the term independent of x in the expansion of $(2 + 3x - 4x^2)\left(2 - \frac{1}{x}\right)^3$. 3

(c) It is known that at noon the sun is hidden by clouds on an average of two days out of every three. If 5 consecutive days are taken, find the probability of the sun shining at noon on

- (i) each day
- (ii) the first 4 days only
- (iii) 4 of the days
- (iv) at least 4 days 8

Question 9 (19 marks)

(a) The points $T(2at, at^2)$ and $P(2ap, ap^2)$ lie on the parabola $x^2 = 4ay$. The equation of the normal to the parabola at T is $x + ty = 2at + at^3$.

(i) Show that the normal at T and P intersect at the point W with co-ordinates $(-apt(p+t), a(t^2 + tp + p^2 + 2))$. 4

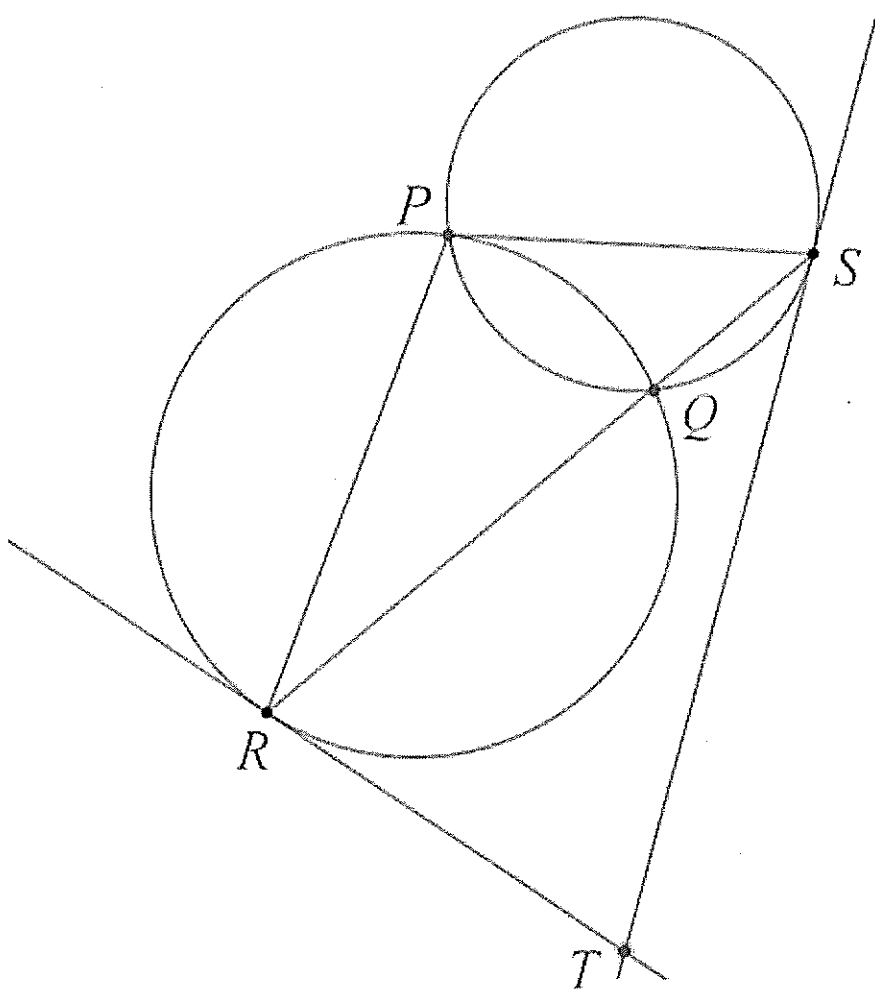
(ii) The equation of the chord TP is $y = \frac{(p+t)x}{2} - apt$. If the chord PT passes through $(0, a)$, show that $pt = -1$. 2

(iii)

(α) If the chord passes through $(0, a)$, show that the equation of the locus of W is a parabola. 4

(β) What is the focal length of this parabola? 1

(b) The circles intersect at P and Q . RQS is a straight line. TR and TS are tangents.



(i) Copy or trace the diagram into your writing booklet, including a construction line from P to Q .

1

(ii) Prove that $PRTS$ is a cyclic quadrilateral.

4

(c) Solve $3\sin\theta + \cos\theta = 2$ by using $t = \tan\frac{\theta}{2}$, $0 \leq \theta \leq 2\pi$.

3

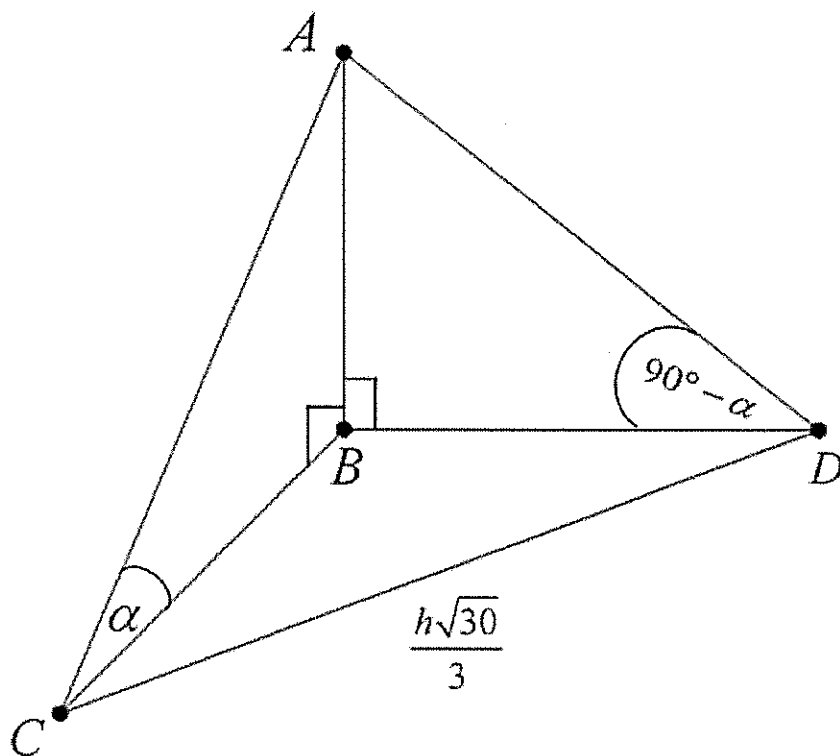
Question 10 (12 marks)

- (a) When the polynomial $P(x)$ is divided by $x+1$, the remainder is 6 and when it is divided by $x-3$, the remainder is -2. Find the remainder when $P(x)$ is divided by $x^2 - 2x - 3$. 4

- (b) Charles is at point C south of a tower AB of height h metres. His friend Daniel is at a point D , which is closer to the tower and east of it. The angles of elevation of the top A of the tower from Charles and Daniel's positions are α and $90^\circ - \alpha$ respectively.

The distance CD between Charles and Daniel is $\frac{h\sqrt{30}}{3}$ metres.

- (i) Show that $3 \tan^4 \alpha - 10 \tan^2 \alpha + 3 = 0$. 5
- (ii) Find α by solving the equation given in (i) 3



End of Examination

Please remember to write 'End of Solutions' on your answer paper.

12 Extension 1 Half Yearly 2016 Solutions

1 A 2 A 3 D 4 B 5 D

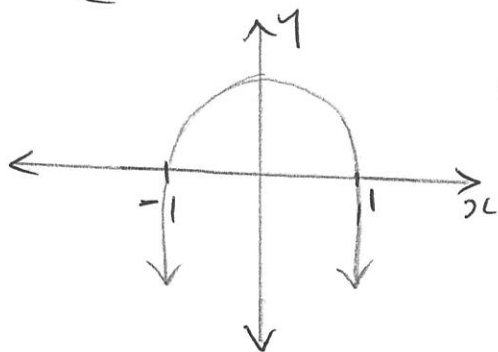
Question 6 (19 marks)

(a) $\frac{2x}{x+1} \leq 1$

$$(x+1)^2 \times \frac{2x}{x+1} \leq (x+1)^2, x \neq -1$$

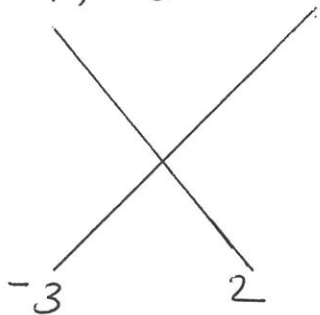
$$(x+1)2x \leq (x+1)^2$$

$$(x+1)(1-x) \geq 0$$



$$\underline{\underline{-1 < x \leq 1}}$$

(b) A(-4, -6) B(6, -1)



$$x = \frac{(-3 \times 6) + (2 \times -4)}{-3 + 2} = 26$$

$$y = \frac{(-3 \times -1) + (2 \times -6)}{-3 + 2} = 9$$

$$\underline{\underline{P(26, 9)}}$$

(c) $\log_e(10x+24) = 2 \log_e x$

$$\log_e(10x+24) = \log_e x^2$$

$$e^{\log_e(10x+24)} = e^{\log_e x^2}$$

$$10x+24 = x^2$$

$$x^2 - 10x - 24 = 0$$

$$(x-12)(x+2) = 0$$

$$x = 12 \text{ or } -2$$

$$\underline{\underline{x = 12}} \quad (\because \log_e x > 0)$$

(d)(i)

$$\text{Let } \sin \alpha + \sqrt{3} \cos \alpha = A \sin(\alpha + \alpha)$$

$$= A \sin \alpha \cos \alpha + A \cos \alpha \sin \alpha$$

Equating coefficients of $\sin \alpha$ and $\cos \alpha$

$$1 = A \cos \alpha$$

$$\sqrt{3} = A \sin \alpha$$

$$A^2 = 4$$

$$A = 2 \quad (\because A > 0)$$

$$\sin \alpha = \frac{\sqrt{3}}{2}$$

$$\cos \alpha = \frac{1}{2}$$

$$\therefore \alpha = \frac{\pi}{3}$$

$$\sin x + \sqrt{3} \cos x = 2 \sin\left(x + \frac{\pi}{3}\right)$$

$$(ii) 2 \sin\left(x + \frac{\pi}{3}\right) = \sqrt{2}$$

$$\sin\left(x + \frac{\pi}{3}\right) = \frac{1}{\sqrt{2}}$$

$$x = \frac{5\pi}{12}, \frac{23\pi}{12} \quad (2)$$

$$(e) (i) \frac{dy}{dx} = e^{-2x}$$

$$y = \int e^{-2x} dx$$

$$= \frac{e^{-2x}}{-2} + C$$

$$\text{When } x=0, y=1$$

$$1 = -\frac{1}{2} + C$$

$$C = \frac{3}{2}$$

$$\therefore y = \frac{e^{-2x}}{-2} + \frac{3}{2} \quad (3)$$

$$(ii) \text{ when } x = -\frac{1}{2} \log_e 3$$

$$y = \frac{e^{-2 \times -\frac{1}{2} \log_e 3}}{-2} + \frac{3}{2}$$

$$= -\frac{3}{2} + \frac{3}{2} = 0$$

$$\text{Point } \left(-\frac{1}{2} \log_e 3, 0\right)$$

Question 7 (23 marks)

$$(a) (i) y = x^2 \log_e x^2$$

$$y' = x^2 \times \frac{1}{x^2} \times 2x + \log_e(x^2) \times 2x$$

$$= 2x + 2x \log_e(x^2)$$

$$= 2x(1 + \log_e x^2) \quad (3)$$

$$(ii) y = \frac{e^x}{x^3}$$

$$y' = \frac{x^3 \times e^x - e^x \times 3x^2}{x^6} \quad (3)$$

$$= \frac{x^2 e^x (x-3)}{x^6} = \frac{e^x (x-3)}{x^4}$$

$$(iii) y = \log_e \left(\frac{x}{\sqrt{x^2+1}} \right)$$

$$= \log_e x - \log_e \sqrt{x^2+1}$$

$$= \log_e x - \log_e (x^2+1)^{\frac{1}{2}}$$

$$= \log_e x - \frac{1}{2} \log_e (x^2+1)$$

$$y' = \frac{1}{x} - \frac{1}{2} \times \frac{1}{x^2+1} \times 2x \quad (4)$$

$$= \frac{1}{x} - \frac{x}{x^2+1} = \frac{1}{x(x^2+1)}$$

$$(b) (i) \int x^2 e^{x^3+1} dx$$

$$= \frac{1}{3} \int 3x^2 e^{x^3+1} dx$$

$$= \frac{1}{3} e^{x^3+1} + C \quad (3)$$

$$(ii) \int \frac{3x}{x^2+11} dx$$

$$= 3 \int \frac{x dx}{x^2+11}$$

$$= \frac{3}{2} \int \frac{2x dx}{x^2+11} \quad (3)$$

$$= \frac{3}{2} \log_e(x^2+11) + C$$

$$(iii) \int_1^{\log_e 2} \frac{e^{3x} + 1}{e^x} dx$$

$$= \int_1^{\log_e 2} (e^{2x} + e^{-x}) dx$$

$$= \left[\frac{e^{2x}}{2} + \frac{e^{-x}}{-1} \right]_1^{\log_e 2}$$

$$= \left[\frac{e^{2x}}{2} - \frac{1}{e^x} \right]_1^{\log_e 2}$$

page 3

$$= \left(\frac{e^{2 \times \log_e 2}}{2} - \frac{1}{e^{\log_e 2}} \right) - \left(\frac{e^2}{2} - \frac{1}{e} \right)$$

$$= \frac{4}{2} - \frac{1}{2} - \frac{e^2}{2} + \frac{1}{e}$$

$$= \frac{3}{2} + \frac{1}{e} - \frac{e^2}{2}$$

$$(c) (i)$$

$$\frac{d}{dx} [\log_e(x^2-5x+7)] = \frac{2x-5}{x^2-5x+7} \quad (2)$$

$$\int_3^4 \frac{2x-5}{x^2-5x+7} dx = [\log_e |x^2-5x+7|]_3^4$$

$$= \log_e |16-20+7| - \log_e |9-15+7|$$

$$= \log_e |3| - \log_e |1|$$

$$= \log_e 3 = \underline{\underline{1.099}} \quad (2)$$

Question 8 (15 marks)

page 4

$$1 \times 1! + 2 \times 2! + 3 \times 3! + \dots + n \times n! = (n+1)! - 1$$

$$\underline{n=1}$$

$$\text{LHS} = 1 \times 1! = 1$$

$$\text{RHS} = 2! - 1 = 1$$

$$\text{LHS} = \text{RHS} \quad \therefore \text{true for } n=1$$

Assume the result is true for $n=k$

$$1 \times 1! + 2 \times 2! + 3 \times 3! + \dots + k \times k! = (k+1)! - 1 \quad \text{--- (1)}$$

To prove for $n=k+1$

$$1 \times 1! + 2 \times 2! + 3 \times 3! + \dots + k \times k! + (k+1) \times (k+1)!$$

$$= (k+2)! - 1 \quad \text{--- (2)}$$

LHS of (2)

$$1 \times 1! + 2 \times 2! + 3 \times 3! + \dots + k \times k! + (k+1) \times (k+1)!$$

$$= (k+1)! - 1 + (k+1) \times (k+1)!$$

$$= (k+1)! + (k+1)(k+1)! - 1$$

$$= (k+1)! [1 + k+1] - 1 = (k+1)! (k+2) - 1$$

$$= (k+2)! - 1 = \text{RHS of (2)}$$

By the principle of Mathematical Induction
the result is true for $n \geq 1$.

$$(b) \left(2 - \frac{1}{x}\right)^3 = 8 - \frac{12}{x} + \frac{6}{x^2} - \frac{1}{x^3}$$

page 5

$$\left(2 + 3x - 4x^2\right) \left(8 - \frac{12}{x} + \frac{6}{x^2} - \frac{1}{x^3}\right)$$

Terms independent of x

$$16 + 3x \times \frac{-12}{x} - 4x^2 \times \frac{6}{x^2}$$

$$= 16 - 36 - 24 = \underline{\underline{-44}}$$

(3)

$$(c) p = \frac{1}{3} \quad q = \frac{2}{3} \quad n = 5$$

$$(a) P(x=5) = {}^5C_5 \left(\frac{1}{3}\right)^5 \left(\frac{2}{3}\right)^0 = \frac{1}{243} = 0.0041$$

$$(b) P(\text{sun shining at noon on the first 4 days}) = \frac{1}{3} \times \frac{1}{3} \times \frac{1}{3} \times \frac{1}{3} \times \frac{2}{3} = \frac{2}{243} = 0.0082$$

$$(c) P(x=4) = {}^5C_4 \left(\frac{1}{3}\right)^4 \left(\frac{2}{3}\right)^1 = \frac{10}{243} = 0.0412$$

(8)

$$(d) P(x=4) + P(x=5) = \frac{10}{243} + \frac{1}{243} = \frac{11}{243} = 0.0453$$

Question 9 (19 marks)

$$(a)(i) x + ty = 2at + at^3 \text{ --- (1)}$$

$$x + py = 2ap + ap^3 \text{ --- (2)}$$

$$\textcircled{1} - \textcircled{2} \quad ty - py = 2a(t-p) + a(t^3 - p^3)$$

$$y(t-p) = 2a(t-p) + a(t-p)(t^2 + tp + p^2)$$

$$y = 2a + a(t^2 + tp + p^2) \\ = a(t^2 + tp + p^2 + 2)$$

$$x = 2at + at^3 - ty \\ = 2at + at^3 - at(t^2 + tp + p^2 + 2) \\ = 2at + at^3 - at^3 - apt^2 - ap^2t - 2at \\ = -apt^2 - ap^2t = -apt(t+p) \quad (4)$$

$$\therefore W(-apt(t+p), a(t^2 + tp + p^2 + 2))$$

(ii) substitute (0, a) in $y = \frac{1}{2}(p+t)x - apt$

$$a = 0 - apt \\ 1 = -pt \quad \therefore \underline{pt = -1} \quad (2)$$

(iii) (a) $x = -apt(p+t)$

$$y = a(t^2 + tp + p^2 + 2)$$

substitute $tp = -1$

$$x = a(p+t) \quad \text{--- (1)}$$

$$y = a(t^2 + p^2 + 1) \quad \text{--- (2)}$$

squaring (1) $(p+t)^2 = \frac{x^2}{a^2}$

$$\begin{aligned}
 y &= a(t^2 + p^2 + 2pt - 2pt + 1) \\
 &= a[(t+p)^2 - 2pt + 1] \\
 &= a[(t+p)^2 + 3] \\
 &= a\left[\frac{x^2}{a^2} + 3\right] = \frac{x^2}{a} + 3a
 \end{aligned}$$

$$\frac{x^2}{a} = y - 3a$$

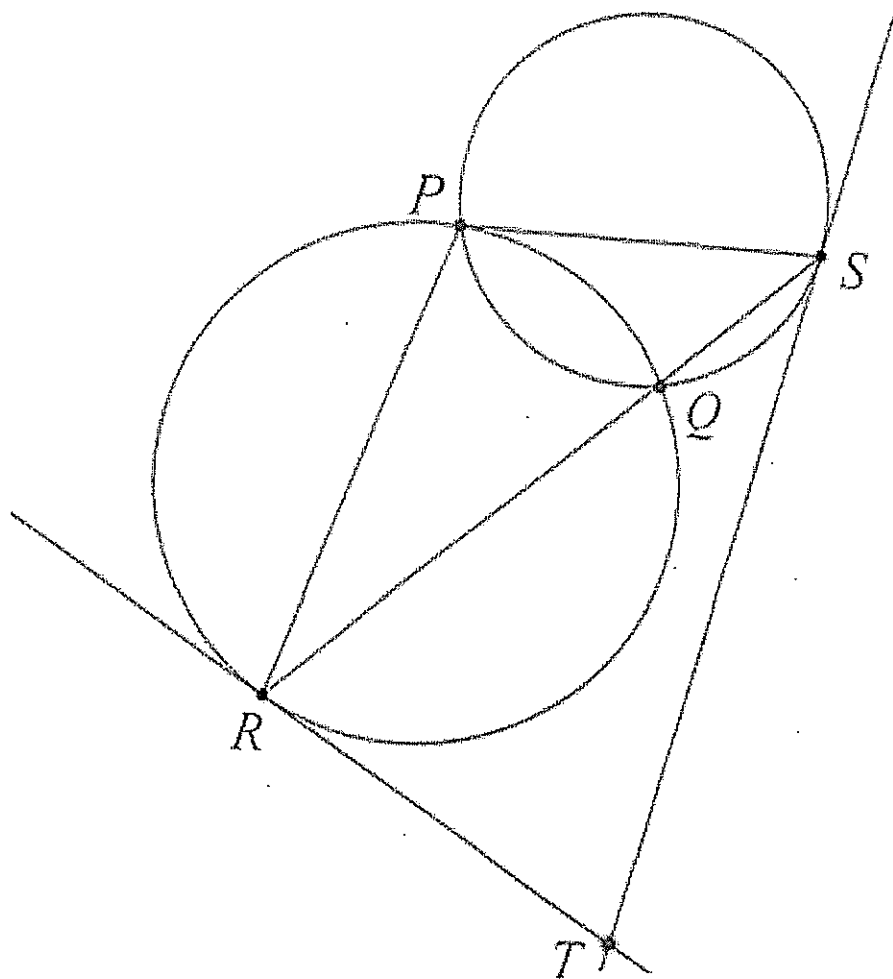
(4)

$$\underline{\underline{x^2 = a(y - 3a)}}$$

$$(p) \quad x^2 = 4 \times \frac{a}{4} (y - 3a)$$

$$\text{Focal length} = \underline{\underline{\frac{a}{4}}}$$

(1)



①

Let $\angle TSQ = x$

$\angle SPQ = x$ (angle between tangent and chord is equal to angle in the alternate segment)

Let $\angle TRQ = y$

$\angle RPQ = y$ (alternate segment theorem)

$\angle RTS = 180 - (x + y)$ (angle sum of $\triangle RTS$)

$\angle RTS + \angle RPS = 180 - (x + y) + x + y = 180$

PRTS is a cyclic quadrilateral (4)
(opposite angles of a quadrilateral are supplementary)

page 1

$$(c) 3 \sin \theta + \cos \theta = 2$$

$$\text{Substitute } \sin \theta = \frac{2t}{1+t^2} \quad \text{and } \cos \theta = \frac{1-t^2}{1+t^2}$$

$$\frac{3 \times 2t}{1+t^2} + \frac{1-t^2}{1+t^2} = 2$$

$$\frac{6t + 1 - t^2}{1+t^2} = 2$$

$$3t^2 - 6t + 1 = 0$$

$$t = \frac{6 \pm \sqrt{24}}{6}$$

(3)

$$\frac{\tan \theta}{2} = \frac{6 \pm \sqrt{24}}{2}$$

$$\theta = \underline{\underline{0.3630^\circ, 2.1351^\circ}}$$

Question 10 (12 marks)

(a) Let $ax+b$ be the remainder

$$p(x) = (x-3)(x+1)Q(x) + ax+b$$

$$p(-1) = 6, \quad p(3) = -2$$

$$3a+b = -2$$

$$-a+b = 6$$

$$\underline{4a = -8}$$

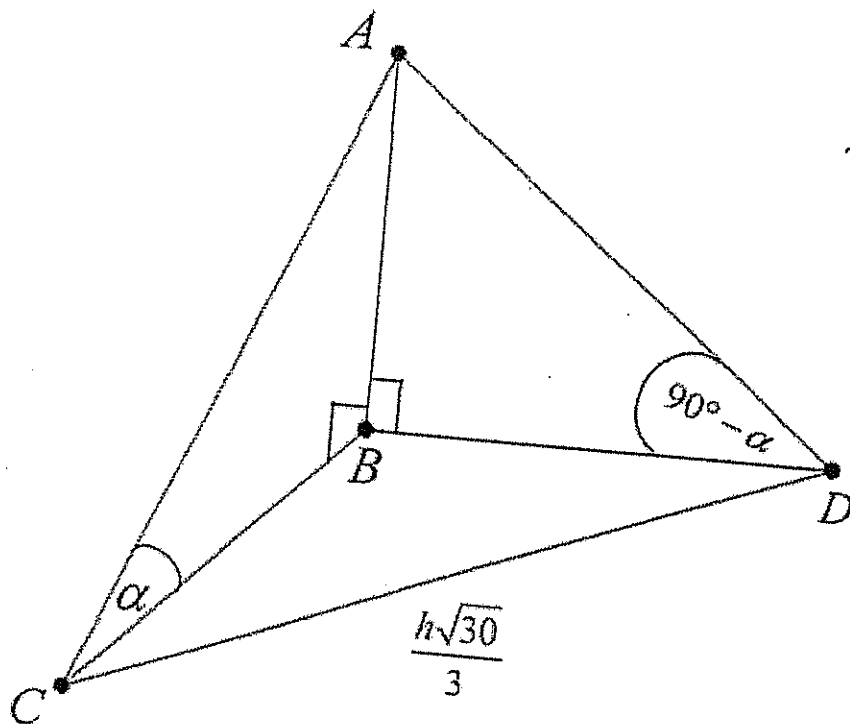
$$a = -2$$

$$b = 6+a$$
$$= 4$$

(4)

$$\text{remainder} = \underline{\underline{-2x+4}}$$

(b)



$$\tan \alpha = \frac{h}{BC}$$

$$BC = \frac{h}{\tan \alpha}$$

$$\tan(90^\circ - \alpha) = \frac{h}{BD}$$

$$BD = \frac{h}{\tan(90^\circ - \alpha)}$$

Apply Pythagoras' theorem in $\triangle BCD$

$$\frac{h^2}{\tan^2 \alpha} + \frac{h^2}{\tan^2(90^\circ - \alpha)} = \frac{h^2 \times 30}{9}$$

$$h^2 \left[\frac{1}{\tan^2 \alpha} + \frac{1}{\tan^2(90^\circ - \alpha)} \right] = \frac{30h^2}{9}$$

$$\frac{1}{\tan^2 \alpha} + \frac{1}{\tan^2(90^\circ - \alpha)} = \frac{10}{3}$$

$$\frac{1}{\tan^2 \alpha} + \frac{1}{\frac{1}{\tan^2 \alpha}} = \frac{10}{3} \quad \left(\because \tan(90^\circ - \alpha) = \frac{1}{\tan \alpha} \right)$$

$$\frac{1}{\tan^2 \alpha} + \tan^2 \alpha = \frac{10}{3}$$

$$1 + \tan^4 \alpha = \frac{10 \tan^2 \alpha}{3}$$

$$3 + 3 \tan^4 \alpha = 10 \tan^2 \alpha$$

$$\therefore 3 + 3 \tan^4 \alpha = 10 \tan^2 \alpha$$

$$3 \tan^4 \alpha - 10 \tan^2 \alpha + 3 = 0$$

(5)

$$\text{Let } m = \tan^2 \alpha$$

$$3m^2 - 10m + 3 = 0$$

$$(m-3)(3m-1) = 0$$

$$m = 3 \text{ or } m = \frac{1}{3}$$

$$\tan^2 \alpha = 3 \text{ or } \tan^2 \alpha = \frac{1}{3}$$

$$\tan \alpha = \pm \sqrt{3} \text{ or } \tan \alpha = \pm \frac{1}{\sqrt{3}}$$

(3)

$$\underline{\underline{\alpha = 30^\circ \text{ or } 60^\circ}}$$

